

**IN THE UNITED STATES DISTRICT COURT
FOR THE DISTRICT OF DELAWARE**

QORVO, INC.,	)	
	)	
Plaintiff,	)	
	)	Case No. 1:21-cv-01417-JPM
v.	)	
	)	
AKOUSTIS TECHNOLOGIES, INC. and	)	
AKOUSTIS, INC.,	)	
	)	
Defendants.	)	

CLAIM CONSTRUCTION ORDER

This case is before the Court for claim construction pursuant to Markman v. Westview Instruments, Inc., 52 F.3d 967, 979 (Fed. Cir. 1995), aff'd, 517 U.S. 370 (1996). A Markman hearing was held on November 11, 2023. (ECF No. 113.) Present were Robert Masters, Trevor Quist, Jeremy Tigan, Jonathan DeFosse, and Timothy Cremen, counsel for Plaintiff, and David Jakopin, Robert Fuhrer, Dianne Sweeney, Stephen Brauerman, and Theresa Roozen, counsel for Defendants. (Id.)

I. BACKGROUND

This is a civil action for patent infringement of one or more of the claims of U.S. Patent No. 7,522,018 (the “’018 Patent”), U.S. Patent No. 9,735,755 (the “’755 Patent”), and U.S. Patent No. 10,256,786 (the “’786 Patent”) (collectively, “the asserted patents”). (ECF No. 125.) Plaintiff Qorvo, Inc. (“Qorvo”) claims that Defendants Akoustis Technologies, Inc. and Akoustis, Inc.

(collectively, “Akoustis”)¹ infringe Qorvo’s patents with its bulk acoustic wave (BAW) resonator filters. (Id.) Qorvo further alleges violations of additional laws, and in total Qorvo alleges that Akoustis engaged in

patent infringement, false advertising, false patent marking, unfair competition, trade secret misappropriation, racketeering, and civil conspiracy, respectively arising under the patent laws of the United States, 35 U.S.C. § 1, et seq., the Lanham Act, 15 U.S.C. § 1125, the Defense of Trade Secrets Act (“DTSA”) (18 U.S.C. § 1832, et seq.), the Racketeer Influenced and Corrupt Organizations Act (“RICO”) (18 U.S.C. § 1961, et seq.), the North Carolina Trade Secrets Protection Act (N.C. Gen. Stat § 66-152, et seq.), and civil conspiracy laws of the state of North Carolina.

(Id. ¶ 1.)

A. Factual Background

Plaintiff Qorvo, Inc. (“Qorvo”) is a Delaware corporation with its principal place of business in Greensboro, North Carolina. (ECF No. 28 ¶ 2.) Defendants are Delaware corporations with its principal place of business in Huntersville, North Carolina. (Id. ¶¶ 3–4.) The Parties are competitors in the bulk acoustic wave (“BAW”) filter market. (Id. ¶ 9.) BAW filters “are a vital component in advanced frequency filtering solutions for wireless devices, as well as the world’s most advanced radar and communications systems.” (Id. ¶ 15.) “[W]ireless devices communicate in particularized frequency bands in the electromagnetic spectrum that are assigned by government entities to prevent interferences between different devices and service providers.” (Id.) As a result, “it is important to filter out signals, through BAW filters, from those frequencies that are irrelevant to a particular device (i.e. communications in frequency bands above or below that assigned to the

¹ While “Akoustis” refers to two entities, the Parties generally treat these entities as if they operate as one single entity, and oftentimes refer to them as a singular noun. (See, e.g., ECF No. 108 at PageID 1951, 1966 (“Akoustis offers” and “Akoustis fails to provide”).) For the sake of consistency, this Order will refer to Akoustis as if it was a singular entity throughout.

device) to improve communications efficacy.” (Id.) Essentially, wireless devices communicate on certain designated frequency channels, and BAW filters serve to reduce interference from other channels that would make such communication difficult.

The main portion of a BAW filter consists of “a piezoelectric² layer sandwiched between two metallic electrodes.” (ECF No. 100 at PageID 895.) By passing electric currents at a specific frequency through the metal electrodes and piezoelectric layer, the BAW filter will vibrate and create a standing acoustic wave. (Id. at PageID 865–96; ECF No. 113 at PageID 2205–07.) This standing wave results in the BAW filter’s filtering effect, and allows only desired frequencies to pass. (ECF No. 100 at PageID 896.) The electrodes and piezoelectric layers are mounted on a substrate, which serves to reduce leakage of the acoustic waves. (Id.) This substrate may take the form of various designs such as a Film Bulk Acoustic Resonator (FBAR), which uses an air cavity as a reflector, or a Solidly Mounted Resonator (BAW-SMR or SMR), which uses solid material as the reflector. (Id.) Certain designs may also seek to reduce lateral leakage of the acoustic waves to improve efficiency of the device. (Id.)

B. Procedural Background

Qorvo filed its Complaint on October 4, 2021. (ECF No. 1.) Qorvo filed its First Amended Complaint on February 18, 2022. (ECF No. 28.) Qorvo filed its Second Amended Complaint, with leave of the Court, on February 8, 2023. (ECF No. 125.)

Qorvo filed its Claim Construction Opening Brief and associated Declaration on August 30, 2022. (ECF Nos. 100, 101.) Akoustis filed its Claim Construction Answering Brief and

² Piezoelectric materials are materials that deform (i.e., change shape) when an electric voltage is applied across the material. (ECF No. 113 at PageID 2204–05.) When the voltage is removed, the material returns to its original shape. (Id.) This property also works similarly in reverse, and piezoelectric materials will produce a voltage in response to applied deformation. (Id.)

associated Declaration on September 28, 2022. (ECF No. 108, 109.) Akoustis filed its Claim Construction Opening Brief and associated Declaration on August 30, 2022. (ECF Nos. 102, 103.) Qorvo filed its Claim Construction Answering Brief and associated Declaration on September 28, 2022. (ECF Nos. 110, 111.) The Parties filed a Joint Claim Construction and Prehearing Statement on October 7, 2022. (ECF No. 112.) The Court held a Claims Construction Hearing on November 21, 2022. (ECF No. 113.)

C. The Patents-in-Suit

The ‘018 Patent describes a resonator with a layer structure consisting of a piezoelectric layer and a top and bottom electrode layer, in which the top electrode layer is thinner than the bottom electrode layer. (ECF No. 28-1 at PageID 326.) There is no disagreement between the Parties as to the definition of claim terms in this patent.

The ‘755 Patent describes a resonator similar to that in the ‘018 Patent, but with the addition of a passivation layer on the top of the top electrode as well as a material in the outer region of the resonator. (Id. at PageID 336.) These additions serve to reduce lateral leakage of mechanical energy from the active region of the resonator. (Id.)

The ‘786 Patent describes an acoustic resonator device. (ECF No. 11-1 at PageID 112.) There are a number of features of the resonator described in the patent, but the main portions at issue in the instant litigation are those involving references to single crystal materials and the piezoelectric layer. (See generally, id.)

Qorvo is the assignee of the ‘018 Patent and ‘755 Patent, and Akoustis is the assignee of the ‘786 Patent. (ECF No. 28 ¶¶ 27, 32; ECF No. 11-1 at PageID 112.)

II. LEGAL STANDARD

“It is a ‘bedrock principle’ of patent law that ‘the claims of a patent define the invention to which the patentee is entitled the right to exclude.’” Phillips v. AWH Corp., 415 F.3d 1303, 1312 (Fed. Cir. 2005) (quoting Innova/Pure Water, Inc. v. Safari Water Filtration Systems, Inc., 381 F.3d 1111, 1115 (Fed. Cir. 2004)). Courts, as a matter of law, must construe the claims of a patent in order to ascertain precisely what it is that is patented. See id.; see also Markman v. Westview Instruments, Inc., 517 U.S. 370, 387 (1996).

In engaging in that exercise, the words in the claims are “generally given their ordinary and customary meaning,” that is, “the meaning that the term would have to a person of ordinary skill in the art in question at the time of the invention.” Phillips, 415 F.3d at 1312–13 (internal citations and quotation marks omitted). This ordinary and customary meaning “may be readily apparent even to lay judges,” and where that is the case, claim construction involves “little more than the application of the widely accepted meaning of commonly understood words.” Id. at 1314 (citing Brown v. 3M, 265 F.3d 1349, 1352 (Fed. Cir. 2001)).

However, as the ordinary and customary meaning is often not immediately apparent, courts must look to other sources of evidence—“the words of the claims themselves, the remainder of the specification, the prosecution history, and extrinsic evidence concerning relevant scientific principles, the meaning of technical terms, and the state of the art.” Id. (citing Innova, 381 F.3d at 1116). In Phillips, the United States Court of Appeals for the Federal Circuit provided guidance on the relative weight given to evidence from these various sources. Id.

First, “the claims themselves provide substantial guidance as to the meaning of particular claim terms,” particularly the “context in which a term is used in the asserted claim.” Id. at 1314. But because claims are also part of a “fully integrated written instrument,” they must “be read in

view of the specification, of which they are a part.” Markman, 52 F.3d at 978, 979 (citations omitted). As the Federal Circuit has stressed, “[a] patent’s specification provides necessary context for understanding the claims, and is always highly relevant to the claim construction analysis.” Abbott Labs. v. Sandoz, Inc., 566 F.3d 1282, 1288 (Fed. Cir. 2009) (en banc in part) (quoting Phillips, 415 F.3d at 1315). Further, “sometimes the specification offers practically incontrovertible directions about claim meaning,” as when inventors “act as their own lexicographers and give a specialized definition of claim terms,” or “intentionally disclaim, or disavow, subject matter that would otherwise fall within the scope of the claim.” Id. (internal citations and quotation marks omitted). But the Court must take care neither “to import limitations into the claims from the specification,” nor to allow “the claims to enlarge what is patented beyond what the inventor has described as the invention.” Id. at 1288 (internal citations and quotation marks omitted). In addition, “a particular embodiment appearing in the written description may not be read into a claim when the claim language is broader than the embodiment.” Resonate Inc. v. Alteon Websystems, Inc., 338 F.3d 1360, 1364–65 (Fed. Cir. 2003).

The prosecution history of the patent is the other type of “intrinsic evidence,” along with the specification, courts consider when determining the meaning of disputed terms. Phillips, 415 F.3d at 1317.

Finally, courts may consider extrinsic evidence—that is, “all evidence external to the patent and prosecution history, including expert and inventor testimony, dictionaries, and learned treatises.” Id. (quoting Markman, 52 F.3d at 980). Such evidence, however, is “less significant than the intrinsic record in determining the legally operative meaning of claim language.” Phillips, 415 F.3d at 1317 (internal quotations and citations omitted).

In engaging in a Markman analysis, a court is not required to “repeat or restate every claim term in order to comply with the ruling that claim construction is for the court.” U.S. Surgical Corp. v. Ethicon, Inc., 103 F.3d 1554, 1568 (Fed. Cir. 1997). Rather, “[c]laim construction is a matter of resolution of disputed meanings and technical scope, to clarify and when necessary to explain what the patentee covered by the claims, for use in the determination of infringement.” Id.

III. CLAIM CONSTRUCTION

A. Agreed-Upon Terms

Patent	Term	Agreed Construction
'018	reference numerals	the reference numerals in the asserted claims of the '018 patent are not limiting and are solely for exemplary purposes.

B. Overview of Disputed Claim Terms

Patent	Disputed Term	Qorvo's Proposed Construction	Akoustis's Proposed Construction
'755	active region	plain and ordinary meaning. if a construction is necessary, “active region” is the region of the baw resonator that is electrically driven to provide the resonator functionality.	that region of the device where motional current density is generated through the piezoelectric layer.
'755	outer region	plain and ordinary meaning. if a construction is necessary, “outer region” is the region of the baw resonator that is not electrically driven.	defined as that region outside the active region.

'755	a density of mechanical energy in the outer region	plain and ordinary meaning. if a construction for “density of mechanical energy” is needed, it means the amount of mechanical energy from acoustic waves leaked from the active region, measured over a volume of the outer region.	the density of mechanical energy means a total density over the full volume of the outer region, or otherwise indefinite.
'755	are not excited	plain and ordinary meaning. if a construction for “are not excited” is needed, it means “are not substantively created.”	“[a]re not excited” means that the wavelengths do not exist in the whole volume of the active region, or otherwise indefinite.
'786	piezoelectric layer	a layer of piezoelectric material having single crystal structure.	a layer of piezoelectric material having single or multiple crystal structure.

1. active region / outer region

Patent	Disputed Term	Qorvo's Proposed Construction	Akoustis's Proposed Construction	Court's Construction
'755	active region	plain and ordinary meaning. if a construction is necessary, “active region” is the region of the baw resonator that is electrically driven to provide the resonator functionality.	that region of the device where motional current density is generated through the piezoelectric layer	the region of the resonator that is electrically driven, plus any portions of the resonator that are located above or below the electrically driven regions
'755	outer region	plain and ordinary meaning. if a construction is necessary, “outer region” is the region of the baw resonator that is not electrically driven.	defined as that region outside the active region.	the region of the resonator that is outside of the active region

The term “active region” appears in claims 1, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 15, and 16 of the ’755 Patent. The term “outer region” appears in claims 1, 3, 5, 6, 7, 9, 10, 11, 12, 14, 15, and 16 of the ’755 Patent.

The primary dispute is whether the active region should be limited to the portion of the resonator that is electrically driven, or if it should instead be defined as the region where “motional current density” is generated. The outer region depends on the relevant definition of the active region. Qorvo argues that the terms should be given their plain and ordinary meaning, but argues that if they are not, that they should be defined based on the definitions provided in the specification and claims. (ECF No. 112-1 at PageID 2180.) Qorvo points to a portion of the specification stating that

The active region 34 is the region of the BAW resonator 10 that is electrically driven. In other words, the active region 34 is the region of the BAW resonator 10 consisting of, in this example, the bottom electrode 14, the top electrode 16, the portion of the piezoelectric layer 12 between the bottom and top electrodes 14 and 16, and the portion of the reflector 18 below the bottom electrode 14. Conversely, an outer region 36 of the BAW resonator 10 is a region of the BAW resonator 10 that is not electrically driven (i.e., the area outside of the active region 34).

(ECF No. 100 at PageID 901 (quoting the ’755 Patent col. 1 ln. 51 – col. 2 ln. 3).) Qorvo also points to Figures in the patent specification that illustrate regions considered to be the active region and outer region in a specific embodiment. (Id. at PageID 902.) Qorvo argues that the intrinsic record provides a clear definition of these terms, and that there is no reason and that it would be inappropriate to introduce extrinsic evidence that would be inconsistent with the intrinsic record. (Id. at PageID 904.) Qorvo also argues that Akoustis’s “motional current density” is not a term that is commonly used in the art. (Id. at PageID 905.)

Akoustis argues that the patent specification is unclear and unhelpful. (ECF No. 102 at PageID 1690–92.) Akoustis more specifically argues that the resonator portion is not electrically driven, and would therefore be excluded from Qorvo’s proposed definition for the active region despite being described as part of the active region in the main embodiment. (Id. at PageID 1692.) Akoustis proposes to replace “electrically driven” with the region where “motional current density” is generated, to remain consistent with the ‘755 Patent and to include the reflector portion of the active region. (Id. at PageID 1692–93.) Akoustis points to multiple pieces of extrinsic evidence, such as scientific journal articles and patent applications, in which the active region is defined as the region where motional current is generated and the electrodes do not touch the piezoelectric material. (Id. at PageID 1693–94.)

In its Answering Brief, Qorvo reiterates its argument that the intrinsic record fully defines the terms and that it is inappropriate to reference any extrinsic evidence. (ECF No. 110 at PageID 2073–75.) Qorvo further argues that the “motional current density” term is not commonly used in the art and would not be understood by a POSITA to have referred to the active region. (Id. at PageID 2075.) Qorvo also argues that the extrinsic evidence also supports its proposed definition of defining an active region based on the electrically driven elements. (Id. at PageID 2076.) Qorvo argues that Akoustis’s definition would inappropriately limit the active region’s height, when only the width of the active region, as shown in certain figures of the patent specification, should be limited. (Id. at PageID 2077–78.) Qorvo argues that its proposed definition would be considered by a POSITA to include portions located above and below the “electrically driven” region, and that the lateral positioning of features is what defines them as being inside or outside of the active region. (Id. at PageID 2078–79.)

In its Answering Brief, Akoustis reiterates its argument that Qorvo’s proposed definition would exclude the reflector portion from the active region, even though the specification explicitly includes the reflector portion within the definition of the active region. (ECF No. 108 at PageID 1958–60.) Akoustis argues that this goes against a fundamental principle of claim construction by excluding the reflectors from the active region. (ECF No. 108 at PageID 1958–59.) Akoustis argues that it is well understood in the art that the active region of a reflector is the region in which there is both physical motion and currents. (Id. at PageID 1960.) Akoustis also argues against the “to provide the resonator functionality” portion of Qorvo’s proposed definition. (Id. at PageID 1958.)

Inventors may “act as their own lexicographers and give a specialized definition of claim terms” that may override the default plain and ordinary meaning that a claim term should normally be given. Phillips, 415 F.3d at 1315. When interpreting a definition that an inventor assigns to a term, “it is unlikely that an inventor would define the invention in a way that excluded the preferred embodiment, or that persons of skill in this field would read the specification in such a way.” Vitronics Corp. v. Conceptiontronic, Inc., 90 F.3d 1576, 1584 (Fed. Cir. 1996.) A claim construction such that “a preferred . . . embodiment would not fall within the scope of the patent claim . . . is rarely, if ever, correct.” (Id. at 1584.)

As Qorvo highlights in its brief, the inventors have provided a definition for active region in the ‘755 Patent. (quoting the ‘755 Patent col. 1 ln. 54 – col. 2 ln. 3.) The inventors define the active region as “the region of the BAW resonator 10 that is electrically driven.” (the ‘755 Patent col. 1 ln. 54–55.) The inventors then further add to this definition by specifying that

the active region 34 is the region of the BAW resonator 10 consisting of, in [a prior art] example, the bottom electrode 14, the top electrode 16, the portion of the piezoelectric layer 12 between the bottom and top electrodes 14 and 16, and the portion of the reflector below the bottom electrode 14.

(The ‘755 Patent, col. 1 ln. 55–61.) This same basic language is also used to describe the active region of a described embodiment later in the specification. (The ‘755 Patent, col. 6 ln. 14–20.) This language, as presented in the specification, is structured so as to provide a definition that the active region is the region that is “electrically driven,” and then provide additional information that, in the contemplated embodiments and prior art, this electrically driven region is made up of the bottom and top electrodes, the piezoelectric layer located between the electrodes, and the portion of the reflector below the bottom electrode. (Id., col. 1 ln. 55–61.)

However, an issue arises when trying to adopt this definition standing alone. As described throughout the ‘755 Patent, the reflector is not “electrically driven.” (See generally the ‘755 Patent.) The reflector serves to contain mechanical energy within the active region and “operates to prevent acoustic waves from leaking longitudinally, or vertically, from the BAW resonator 10 into the substrate . . . below the reflector.” (the ‘755 Patent, col. 2 ln. 4–9.) In order to operate the BAW resonator, an electrical signal is applied to the bottom and top electrodes and passes through the piezoelectric layer to cause it to deform. (the ‘755 Patent, col. 1 ln. 51–54.) While the electrodes and portions of the piezoelectric layer are electrically driven during this process, the resonator is not. (see generally the ‘755 patent.) Further, both parties agree that the reflector, as described in the ‘755 patent, is not electrically driven. (ECF No. 113 at PageID 2253–54, 2258.) Similarly, the ‘755 patent also characterizes at least portions of the passivation layer as being located in the active region, despite the passivation layer being neither electrically driven nor specifically recited as being in the active region. (E.g., the ‘755 Patent, col. 6 ln. 12–14 (“a thickness (T_{pa}) of the passivation layer 60 is the same in both the active region 62 and in the outer region 64”).)

Because of this conflict, a construction of the active region that is limited to the electrically driven portions would improperly exclude the portion of resonator below the electrodes, which is explicitly described as being part of the active region. Conversely, a construction that includes the resonator would conflict with the definition stating that the active region is limited to the electrically driven portions, since the resonator is not electrically driven. For inventors to serve as their own lexicographers, “any special definition given to a word must be clearly defined in the specification.” Markman, 52 F.3d at 980 (citing Intellicall, Inc. v. Phonometrics, Inc., 952 F.2d 1384, 1388 (Fed. Cir. 1992)). Despite attempting to act as their own lexicographers, the inventors have provided a definition for the term which is not clearly defined because it is inconsistent with itself. It is therefore necessary to evaluate this term in view of the specification to determine the appropriate construction for this term. Id. at 979 (“Claims must be read in view of the specification, of which they are a part.”).

Throughout the figures in the specification, the active region is consistently illustrated, using dashed lines, as being a region extending above and below the top electrodes. (See generally the ‘755 Patent.) These dashed lines illustrate the boundaries of the electrically driven portions of the device, as well as any substrate, including any reflector or air cavity, upon which the electrically driven portions are mounted and any passivation layers or border rings that are deposited on the top electrode. (Id.) It also appears that only the portion of the piezoelectric layer through which electricity is passed would be part of the active region, because each figure illustrates the portions of the piezoelectric layer which are not between the electrodes as being located in the outer region. (Id.) This is consistent with the text of the specification as well, which refers only to the portions of the piezoelectric layer that are located between the electrodes as being part of the active region. (Id., col. 1 ln. 58–59.) Similarly, the figures show that portions of the

reflector located below the bottom electrode and portions of the passivation layer and border ring located above the top electrode are within the active region, but that portions of these features located outside the boundaries of the electrode are in the outer region. (See, e.g., id., figs. 2A, 2B.)

In reading the specification as a whole, it becomes clear that the inventors intended the active region to be defined as the region of the resonator that is electrically driven, plus any portions of the resonator that are located above or below the electrically driven regions.

While the Parties have put forth different suggested constructions for the term “outer region,” it appears that there is not a significant dispute regarding this term, and both parties essentially argue for a definition that is essentially the region of the resonator that is not the active region. (ECF No. 112 at PageID 2180, 2185; ECF No. 113 at PageID 2245.) The only disagreements in the definition of outer region stem directly from the dispute regarding the definition of active region. (ECF No. 112 at PageID 2180, 2185; ECF No. 113 at PageID 2245.)

The inventors have provided a definition for an outer region in the specification, which defines the outer region as “the region of the reference BAW resonator 38 that is not electrically driven or, in other words, the region of the reference BAW resonator 38 that is outside of the active region 62.” (The ‘755 Patent, col. 6 ln. 23–25.) Because this definition is derived from the definition of the electrically driven region, it suffers from a similar inconsistency as the term “active region.” However, when evaluated in view of the specification, it becomes clear that outer region should be defined as the region of the resonator that is outside of the active region. This construction is consistent with the definition that the inventors attempted to provide, which describes the outer region essentially as the region that is outside the active region. (See id.) This construction is also consistent with the figures in the specification, which consistently illustrate

the outer region as being the portion of the resonator that is outside of the active region. (See generally id.)

2. a density of mechanical energy in the outer region

Patent	Disputed Term	Qorvo's Proposed Construction	Akoustis's Proposed Construction	Court's Construction
'755	a density of mechanical energy in the outer region	plain and ordinary meaning. if a construction for “density of mechanical energy” is needed, it means the amount of mechanical energy from acoustic waves leaked from the active region, measured over a volume of the outer region.	the density of mechanical energy means a total density over the full volume of the outer region, or otherwise indefinite	the density of mechanical energy over the full volume of the outer region

The term “a density of mechanical energy in the outer region” appears within the larger phrase “a density of mechanical energy in the outer region of the BAW resonator is reduced as compared to a density of mechanical energy in the outer region of the BAW resonator when n is equal to 1” in claims 1 and 11 of the '755 Patent.

The primary dispute in this term is whether the density of mechanical energy must be measured over the entire volume of the outer region, or whether it may instead be measured over smaller volumes. Qorvo argues that “the inventive structure reduces the amount of mechanical energy created by acoustic waves in the active region and leaked laterally into the outer region.” (ECF No. 100 at PageID 907-08.) Qorvo argues that the '755 Patent “does not require the density of mechanical energy to be measured over a particular extent of the volume of the outer region.” (*Id.* at PageID 908.) Qorvo argues that this density measurement could be taken over many different sample volumes of the outer region, and that it is not necessary to measure it over the entire volume of the outer region. (*Id.* at PageID 908-09.) Qorvo argues that Akoustis’s proposed definition is not supported by the intrinsic record, and that Akoustis is attempting to add in a limitation. (*Id.* at PageID 909.) Qorvo notes that “[w]hile Akoustis’[s] construction encompasses

one possible method of comparing energy leakage . . . , it is not the only way to do so.” (Id. at PageID 909.)

Akoustis argues that, since energy density is defined as an amount of energy per unit volume, that this definition requires a defined volume over which the amount of energy is to be measured. (ECF No. 102 at PageID 1695.) Akoustis argues that “what matters generally to a designer, is the total energy lost to the surroundings, not just energy lost in one or more undefined smaller region(s), without knowing what happens in other undefined smaller region(s).” (Id.) Akoustis argues that the only way to measure this total loss is by comparing the energy in the entire outer region, and not just in various specific regions. (Id. at PageID 1695.) Akoustis notes that there may be situations in which the total energy in the outer region increases and the quality factor (Q) decreases, but energy in a particular smaller volume might show a reduced energy density. (Id. at PageID 1696.) Akoustis further argues that Qorvo’s proposed definition would render the claim indefinite and therefore invalid because “a volume” is not defined. (Id. at PageID 1696.)

In its Answering Brief, Qorvo argues that a POSITA would have recognized that smaller volumes could be compared to analyze whether the density of mechanical energy had changed for different n values³. (ECF No. 110 at PageID 2080.) Qorvo argues that the ‘755 Patent does not require that the density of mechanical energy must be measured over the full volume, and that it is improper to add in this limitation. (Id.) Qorvo also argues that Akoustis’s construction is a circular definition, because it uses the word “density” to define “density of mechanical energy”

³ The n value is the ratio of the thickness of the material on the second surface of the piezoelectric layer adjacent to the second electrode in the outer region of the resonator to the thickness of the passivation layer. (The ‘755 Patent, abstract.) In other words, the thickness of the material layer 94 is n times the thickness of the passivation layer 88.

without actually defining density alone. (Id. at PageID 2083.) Qorvo also disagrees with Akoustis's assertion that total energy lost is what really matters, and notes that the intrinsic record shows that the comparison between two density values is what matters in the context of the '755 patent, and not the overall energy. (Id. at PageID 2084.) Qorvo argues that its construction would not render the claims indefinite, because a POSITA would have understood that different volumes could be used for the comparison. (Id.)

In its Answering Brief, Akoustis argues that the volume must be specified, because "too small a volume clearly will not indicate whether there is actual energy reduction in the outer region adequate to improve the overall Q of the device." (ECF No. 108 at PageID 1962.) Akoustis argues that the only unambiguous volume that aligns with improving the Q factor is the full volume. (Id. at PageID 1966.) Akoustis also points to testimony by Qorvo's expert Dr. Bravman, which Akoustis argues supports its proposed construction. (Id. at PageID 1963–65.)

Unlike the previously discussed terms, the inventors do not propose a definition for how the term "a density of mechanical energy in the outer region" should be construed. Throughout the specification, the inventors highlight that it is an important goal to manipulate the n value of the device such that "lateral leakage of mechanical energy from an active region of the BAW resonator is reduced." (E.g., the '755 Patent, abstract.) The '755 Patent teaches that a high Quality Factor (Q) of the device is desirable, and that lateral leakage of mechanical energy from the active region into the outer region reduces Q. (Id., col. 1 ln. 21–22, col. 6 ln. 26–31.) While the claims evaluate a reduction in leakage by comparing the density of mechanical energy in the outer region for different n values, the specifications consistently refer instead simply to "lateral leakage" of mechanical energy from the active region into the outer region. (See generally id.) The only time that the specification refers to a density of mechanical energy is when using it to illustrate a

reduction of lateral leakage for different n values. (Id. col. 6 ln. 58–63, figs. 3A, 3B.) Read in context, this description and its associated figures serve to provide a visual illustration of reduced total energy leakage, which is simply illustrated using density for the purpose of visualization.

Read in view of the specification, the “density of mechanical energy in the outer region” as described in the claims can be better understood as referring to the leakage of mechanical energy from the active region to the outer region.

While Qorvo is correct that a density can generally be measured over a particular smaller sample volume, doing so would be inconsistent with the ‘755 Patent’s goal of reducing overall lateral leakage of mechanical energy. Measuring the density of mechanical energy at certain sample points would allow a comparison of the density of energy at certain points, but this would not provide any useful information regarding overall lateral leakage. In certain circumstances, the mechanical energy density at certain selected points in the outer region might be reduced despite there being an increase in the mechanical density at other non-measured points and an increased lateral leakage overall. The only way that density can be used to evaluate the amount of mechanical energy that escapes through lateral leakage, as described by the specification, is by measuring the density of mechanical energy over the entire volume of the outer region.

3. are not excited in the active region

Patent	Disputed Term	Qorvo's Proposed Construction	Akoustis's Proposed Construction	Court's Construction
'755	are not excited in the active region	plain and ordinary meaning. if a construction for “are not excited” is needed, it means “are not substantively created.	“[a]re not excited” means that the wavelengths do not exist in the whole volume of the active region, or otherwise indefinite	plain and ordinary meaning

The term “are not excited in the active region” appears within the larger phrase “acoustically matched in such a manner that one or more wavelengths that cause energy leakage into the outer region are not excited in the active region” in claims 3 and 12 of the '755 Patent.

The primary disagreement between the Parties is whether “are not excited” means that the wavelengths at issue are not created at all and therefore do not exist in the active region, or whether instead the active region may produce some, though not a substantive amount, of these wavelengths. Qorvo argues that “the intrinsic record in its entirety indicates that the recited non-excitation relates to reducing the lateral leakage of mechanical energy into the outer region.” (ECF No. 100 at PageID 911.) Qorvo argues that wavelengths which are not substantively excited do not cause such leakage, and therefore may occur within the scope of the claim. (*Id.* at PageID 912.) Qorvo argues that Akoustis’s construction would result in a “scientifically impossible construction” and that this construction cannot be applied without specific intrinsic support, which is not present. (*Id.* at PageID 912–13.)

Akoustis argues that Qorvo’s proposed construction would undo the plain meaning of the term. (ECF No. 102 at PageID 1697.) Akoustis argues that it is a fact that all wavelengths in the

active region, even those that are “not substantially created,” will leak to the outer region. (Id.) Akoustis argues that this construction would leave it up to the complete discretion of a POSITA in determining “how much is enough” leakage. (Id. at PageID 1698.) Akoustis also points to parts of the ‘755 patent that use less absolute language such as “reduce” and notes that extrinsic evidence shows the use of terms such as “reduce” or “suppress” as less absolute descriptions. (Id. at PageID 1698–99.)

In its Answering Brief, Qorvo argues that it is not attempting to rewrite the claim terms, but rather that its proposed construction is based on the intrinsic record. (ECF No. 110 at PageID 2087.) Qorvo argues that the claim element in which “not excited” appears gives guidance to its meaning, and that this claim element “is reciting that the wavelengths that cause energy leakage into the outer region are not excited,” which is also a definition that is confirmed throughout the ‘755 Patent. (Id.) Qorvo also argues against Akoustis’s assertion that its claim construction would render the claim indefinite and argues that its proposal is a straightforward definition that wavelengths are not “substantively created” if they are not excited enough to cause them to leak. (Id. at PageID 2088.)

In its Answering Brief, Akoustis reiterates its argument that Qorvo’s proposed construction would render its claims indefinite. (ECF No. 108 at PageID 1967.) Akoustis also reiterates its argument that any wavelengths produced in the active region, no matter how small, will leak to at least some degree into the outer region and therefore Qorvo’s proposed construction is scientifically impossible. (Id. at PageID 1968.)

It appears that Akoustis is correct in its assertion that Qorvo's construction would result in a physical impossibility, because any wavelengths that are produced in the active region would

leak into the outer region to at least some degree.⁴ However, Akoustis's proposed construction imposes an improper claim limitation since it requires that the wavelengths do not exist at all in the active region, and not simply that they are not created. Such wavelengths could "exist" in the active region without being created there, such as if there is an external source generating these wavelengths. Such a restriction is unsupported by the intrinsic record, which makes no reference to whether the wavelengths exist in the active region, but instead is concerned with whether the wavelengths are excited in the active region. (See generally the '755 patent.) As recited by the specification, the intent is that the undesirable wavelengths causing leakage are not created in the active region. (See generally id.) Because the invention creates wavelengths by exciting them, this definition is reflected in the plain and ordinary meaning.

4. piezoelectric layer

Patent	Disputed Term	Qorvo's Proposed Construction	Akoustis's Proposed Construction	Court's Construction
'786	piezoelectric layer	a layer of piezoelectric material having single crystal structure	a layer of piezoelectric material having single or multiple crystal structure	a layer of piezoelectric material having single or multiple crystal structure

The term "piezoelectric layer" appears in claims 1, 8, 12, and 16 of the '786 Patent.

The Parties dispute whether the term "piezoelectric layer" can refer to a piezoelectric layer with a multiple crystal structure, or it is limited only to a layer with a single crystal structure.

⁴ While it may be theoretically possible to create a resonator that perfectly reflects certain acoustic waves and entirely prevents any leakage of certain wavelengths, this is not what the '755 patent teaches. (See generally the '755 Patent.) Rather, the '755 Patent teaches how to reduce, but not necessarily eliminate, leakage. (Id.) Even if certain wavelengths are only nominally created, and their leakage is substantially reduced, there would still be some small portion that would leak from the active region into the outer region.

Qorvo argues that the ‘786 Patent specification repeatedly and consistently characterizes the piezoelectric layer as being single crystal. (ECF No. 100 at PageID 915.) Qorvo notes that “[f]rom the title, to the abstract and figures, to every single embodiment of the invention, the ‘piezoelectric layer’ is invariably linked to the ‘single crystal’ structure identified in Qorvo’s construction.” (Id.) Qorvo also points to the provisional U.S. Patent Application No. 62/360,904 (the “’904 Provisional”) that the ‘786 Patent claims priority to and argues that it is explicitly focused on single crystal piezoelectric materials. (Id. at PageID 916.) Qorvo also argues that under Federal Circuit precedent, “[w]here the specification makes clear that the invention does not include a particular feature, that feature is deemed to be outside the reach of the claims of the patent, even though the language of the claims . . . might be considered broad enough to encompass the feature in question.” (Id. at PageID 917 (quoting SciMed Life Sys., Inc. v. Advanced Cardiovascular Sys., Inc., 242 F.3d 1337, 1341 (Fed. Cir. 2001)).) Qorvo argues that by repeatedly emphasizing the “single crystal” structure, the ‘786 Patent implicitly excludes “multiple crystal” features. (Id.)

Akoustis argues that its proposed construction is the plain and ordinary meaning of the term as understood by a POSITA. (ECF No. 102 at PageID 1700.) Akoustis argues that the “single crystal” term is in none of the independent claims of the ‘786 patent, and instead exists as an optional feature referring to unrelated features such as the seed substrate. (Id. at PageID 1701.) Akoustis argues that “the inventors knew when to use the single-crystal modifier” when the limitation was intended, and they did not use it to limit the piezoelectric layer. (Id. at 1700-01.) Akoustis also argues that there is no mention that the claim is limited in the prosecution history and there is no disclaimer present in the specification. (Id. at PageID 1701.) Akoustis further argues that a single crystal piezoelectric layer is not part of the “inventive aspect” of the ‘786 Patent, and that “the invention is not predicated on using a single-crystal piezoelectric material,

but instead describes a new way of ‘packaging’ a BAW filter, while using conventional materials, such as single-crystal piezoelectric layers.” (Id. at PageID 1702.) Akoustis argues that while a preferred embodiment may be one that has a single crystal piezoelectric layer, the claim term should not be limited to this preferred embodiment. (Id. at PageID 1703.)

In its Answering Brief, Qorvo argues that it is not necessary for “single crystal” to be in the same claim as the disputed piezoelectric layer term. (ECF No. 110 at PageID 2090.) Qorvo argues that the inventors used “piezoelectric layer” as shorthand for “single crystal piezoelectric layer” and that a POSITA would have understood piezoelectric layer, without the single crystal qualifier, to refer specifically to a single crystal piezoelectric layer in this context. (Id. at PageID 2090–91.) Qorvo argues that the patent application was not rejected during prosecution, and therefore that the prosecution history’s lack of any express disclaimer is meaningless. (Id. at PageID 2092.) Qorvo argues that each of the embodiments includes a single crystal piezoelectric layer. (Id. at PageID 2094.) While Qorvo notes that single crystal piezoelectric materials are conventional, it argues that the ‘786 Patent makes specific use of this conventional material. (Id. at PageID 2095.)

In its Answering Brief, Akoustis argues that Qorvo mischaracterizes the teachings of the ‘786 Patent’s intrinsic record. (ECF No. 108 at PageID 1970.) Akoustis reiterates its argument that a resonator with a piezoelectric layer having a single crystal structure is only a preferred embodiment, and not the only possible option, and disagrees with Qorvo’s assertion that the inventors used the term “piezoelectric layer” as shorthand to refer exclusively to a single crystal piezoelectric layer. (Id. at PageID 1971–72.)

“Absent a clear disavowal in the specification or the prosecution history, the patentee is entitled to the full scope of its claim language.” Home Diagnostics, Inc. v. LifeScan, Inc., 381

F.3d 1352, 1358 (Fed. Cir. 2004.) A disavowal of claim term scope “does not require express statements by the patentee identifying the surrendered claim scope. Rather, it may be implicit, so long as it is sufficiently clear.” Rembrandt Patent Innovations, LLC v. Apple, Inc., 716 F. App’x 965, 972 (Fed. Cir. 2017). “Where the specification makes clear that the invention does not include a particular feature, that feature is deemed to be outside the reach of the claims of the patent, even though the language of the claims, read without reference to the specification, might be considered broad enough to encompass the feature in question.” SciMed, 242 F.3d at 1341.

While the inventors make repeated references to single crystal devices, the inventors did not explicitly or implicitly limit the invention only to single crystal devices. There is no reference anywhere in the claims to a single crystal piezoelectric layer. (The ‘786 Patent, claims 1–20.) The only references to single crystal materials in the claims are dependent claims addressing the seed substrate, and not the piezoelectric layer. (Id., claims 2, 9, 13, 17.) The inventors also make it clear throughout the specification that a device with a single crystal piezoelectric layer is one embodiment, but that it is not the only possible embodiment. (See id., col. 1 ln. 15–18. (“the present invention provides techniques related to bulk acoustic wave resonator devices, single crystal bulk acoustic wave resonator devices, single crystal filter and resonator devices, and the like”).) Had the inventors sought to limit the invention only to single crystal piezoelectric devices, it would have been illogical to list “bulk acoustic wave resonator devices” in addition to “single crystal bulk acoustic resonator devices.” (Id.) The inventors knew the difference between single and multiple crystal piezoelectric layers, and the fact that the claims describe piezoelectric layers generally, without a “single crystal” requirement, demonstrates that the claims were intended to cover both single and multiple crystal structures.

Qorvo argues that a number of cases, and SciMed in particular, support its argument that piezoelectric layers should be limited to single crystal structures. (ECF No. 113 at PageID 2317.) However, the facts of these cases can be distinguished from the instant action. In SciMed, the court held that catheters described in the claims were limited to only coaxial configurations, even though the claim terms were broad enough to also cover dual lumen configurations. SciMed, 242 F.3d 1337. The court placed weight on four main characteristics of the patent in reaching this conclusion: (1) the patent described the invention itself as being made in a coaxial configuration, (2) the patent distinguished the invention from prior art using a dual lumen configuration and highlighted the advantages of the coaxial configuration as described in the patent, (3) the patent described the “present invention” as having a coaxial configuration, and (4) the patent specified that “all embodiments of the present invention contemplated and disclosed” were constructed in a coaxial configuration. Id. at 1342–44. Despite these characteristics being factors to consider in whether a patent includes a disavowal, any one of these characteristics standing alone are insufficient to constitute a disavowal. Thorner v. Sony Comput. Ent. America LLC, 669 F.3d 1362, 1366–67 (Fed. Cir. 2012) (noting that “[m]ere criticism of a particular embodiment encompassed in the plain meaning of a claim term is not sufficient to rise to the level of clear disavowal” and “[i]t is likewise not enough that the only embodiments, or all of the embodiments, contain a particular limitation”).

The ‘786 Patent does not describe the invention in such a limited manner. The ‘786 Patent specifications note that the invention may take the form of a BAW resonator device or a single crystal BAW resonator device, among others, highlighting that the invention is not required to have a single crystal structure. (The ‘786 Patent, col. 1 ln. 15–18.) While the ‘786 patent describes embodiments using single crystal piezoelectric layers, it does not distinguish the invention from

prior art with a multiple crystal structure or teach that a single crystal structure is advantageous over a multiple crystal structure. The '786 patent also does not recite that all embodiments have a single crystal piezoelectric layer, and in fact contemplates embodiments that do not have a single crystal structure. (The '786 Patent, col. 1 ln. 15–18.) Because the facts of the instant case are distinguishable, they do not support a finding that the inventors made an express or implicit disavowal of a multiple crystal piezoelectric layer.

IV. SUMMARY OF CONSTRUCTION

Patent	Disputed Term	Court's Construction
'755	active region	the region of the resonator that is electrically driven, plus any portions of the resonator that are located above or below the electrically driven regions
'755	outer region	the region of the resonator that is outside of the active region
'755	a density of mechanical energy in the outer region of the baw resonator is reduced as compared to a density of mechanical energy in the outer region of the baw resonator when n is equal to 1	the density of mechanical energy over the full volume of the outer region
'755	acoustically matched in such a manner that one or more wavelengths that cause energy leakage into the outer region are not excited in the active region	the density of mechanical energy over the full volume of the outer region
'786	piezoelectric layer	a layer of piezoelectric material having single or multiple crystal structure

SO ORDERED, this 14th day of March, 2023.

s/ Jon P. McCalla
JON P. McCALLA
UNITED STATES DISTRICT JUDGE